

Project Progress Report

Amity School of Engineering & Technology

Project Group ID: 265

Project Area: Data Science (Machine Learning / Recommender Systems)

B.Tech (CSE) – VIII Semester

Mid Project Report: Students Project Progress Report (SPPR) for Even Semester of Session 2025-2026

Project Title: CityMatch: A Hyper-Local Urban Relocation Recommender System using Unsupervised Machine Learning and Geospatial Analytics

Students:

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1. Synopsis & Scope Refinement

The project scope was refined from a generic "Liveability Predictor" to a specialized "Decision Support System" (CityMatch).

Pivot: Shifted from descriptive analytics (visualizing data) to prescriptive analytics (recommending locations).

Objective: To solve the multi-criteria optimization problem of urban relocation by mathematically matching user preference vectors (Budget, Health, Commute) with neighbourhood attribute vectors using AI.

2. Literature Review & Methodology Selection

Conducted a targeted review of Unsupervised Learning techniques for urban segmentation.

Algorithm Selection: Selected K-Means Clustering over Hierarchical Clustering due to its efficiency with medium-sized, high-dimensional datasets.

Recommender Logic: Adopted Content-Based Filtering using Cosine Similarity (Vector Space Model) to handle the "Cold Start" problem, ensuring users get recommendations without needing prior interaction history.

3. Data Sourcing & API Integration

We engineered a multi-source data pipeline to overcome the lack of a single unified urban dataset:

Traffic Intelligence: Integrated TomTom Traffic Flow API to fetch real-time "Current Speed" vs. "Free-Flow Speed" metrics, creating a granular Congestion Index that replaces static distance measurements.

Environmental Data: Utilized Open-Meteo API to fetch 2 years (2023–2025) of hourly PM2.5 and NO2 data, enabling the model to account for seasonal pollution variances (e.g., Winter Smog vs. Monsoon).

Geospatial Infrastructure: Used OSMnx to scrape and compute density scores for Schools, Hospitals, and Metro Stations within a 1km radius of 50+ residential centroids.

4. Data Preprocessing & Feature Engineering

Normalization: Applied Min-Max Scaling to bring disparate units (Rent in ₹, Traffic Index 0-10, AQI 0-500) into a uniform [0, 1] vector space.

Feature Inversion: Implemented an inversion transformation ($f(x) = 1 - x$) for negative attributes (Rent, Pollution, Traffic) to ensure the recommendation engine correctly prioritizes "Lower" values as "Better."

Imputation: Handled missing environmental sensor data using Spatial Interpolation (Inverse Distance Weighting) from the nearest active CPCB station.

5. Modelling & Implementation

Clustering Engine: Successfully implemented K-Means Clustering.

Validation: Used the Elbow Method and Silhouette Analysis to scientifically determine optimal $k=4$, revealing distinct neighbourhood archetypes (e.g., "Affordable Industrial" vs. "Premium Green Zones").

Recommendation Engine: Built the Python logic to calculate the Cosine Similarity between a User's Priority Vector and Neighbourhood Feature Vectors.

Traffic Modelling: Constructed a "Composite Traffic Profile" by sampling TomTom API data during Morning Peak, Evening Peak, and Off-Peak hours to create a robust daily average.

6. Application Development (Dashboard)

Framework: Developed a web-based interactive dashboard using Streamlit.

Features:

Dynamic Sliders: Allows users to set weights for 6 key parameters (Rent, Air, Traffic, Connectivity, etc.).

Real-Time Mapping: Integrated Folium maps to visualize top recommended neighbourhoods with interactive markers.

Live Computation: The backend recalculates vector matches instantly upon user input.

7. Project Progress Table

Week	Phase	Key Activities & Progress
1–2	Architecture Design	Finalized "CityMatch" concept; conducted feasibility studies for Open-Meteo and OSMnx APIs.
3	Data Pipeline	Developed automated scripts for fetching historical pollution data and scraping amenity density.
4	Preprocessing	Merged Geospatial, Environmental, and Economic datasets; applied Min-Max Scaling and Normalization .
5	Traffic Integration	Integrated TomTom API ; scripted real-time congestion sampling and "Time Cost" calculations.
6	ML Implementation	Executed K-Means Clustering ($k=4$); validated model performance using Silhouette Scores .
7	Dashboarding	Built Streamlit UI ; successfully connected Recommender Logic to frontend interactive sliders.

8. Challenges and Solutions

Challenge: Lack of historical traffic data due to high commercial costs.

Solution: Implemented a Sampling Strategy using the TomTom Free Tier, querying the API during peak/off-peak windows to construct a statistically valid "Daily Average" proxy.

Challenge: Subjectivity in "Rent" prices.

Solution: Standardized the dataset by using "Average 2BHK Yield" based on a manual domain survey of the 50 target neighbourhoods.

9. Conclusion & Forward Plan

The core "Intelligence Layer" (Clustering + Recommender) is fully operational and integrated with real-time traffic data.

Current Status: The system successfully ranks neighbourhoods based on complex user queries.

Next Steps:

Implement "Dual-Commute Optimization" for multi-location work commutes.

Conduct rigorous "Persona Testing" to validate recommendation accuracy.

Finalize the Thesis Report.

Student Signature:

Faculty Guide Signature:

